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Simplifying algebraic expressions

Simplifying algebraic expressions involves manipulating and rewriting algebraic terms in a simpler or more compact form without changing their original value. This can apply to linear or non-linear expressions.

Example 3: Simplify $\frac{(7x^{-3})\cdot(5x^2)}{5x^{-4}}$.

To simplify $\frac{(7x^{-3})\cdot(5x^2)}{5x^{-4}}$, follow these steps:

i) Combine the terms in the numerator, considering that in a multiplication when the bases are equal, keep the base and add the exponents ($a^m \cdot a^n = a^{m+n}$):

$$7x^{-3} \cdot 5x^2 = (7 \cdot 5) \cdot (x^{-3} \cdot x^2) = 35 \cdot x^{-1} = \frac{35}{x}$$

ii) Divide by the denominator: Since division with exponents is equivalent to subtracting exponents, we can rewrite $x^{-3} \cdot x^2$ and $x^{-1} \cdot x^{-4}$:

$$\frac{35 \cdot x^{-1}}{5 \cdot x^{-4}} = \frac{35}{5 \cdot x^{-4} \cdot x^1} = \frac{35/5}{x^{-3}}$$

iii) Simplify: After combining, we have the following:

$$7 \cdot x^3 = 7x^3$$

Thus, the simplified expression for $(7x^{-3} \cdot 5x^2) \div (5x^{-4})$ is: $7x^3$.

Example 4: Simplify $\frac{\sqrt{y^3}\cdot\sqrt{y}}{y^{-2}}$

To simplify the expression $\frac{\sqrt{y^3}\cdot\sqrt{y}}{y^{-2}}$, you can use the rules of exponents and radicals to combine and simplify terms.

i) Combine radicals:

Since $\sqrt{y^3} = y^{3/2}$ and $\sqrt{y} = y^{1/2}$, you can multiply these two radicals:

$$\sqrt{y^3} \cdot \sqrt{y} = y^{3/2} \cdot y^{1/2}$$

ii) Add the exponents:

Since the bases are the same (y), you can add the exponents:

$$y^{3/2+1/2} = y^2$$

iii) Rewrite the division:
Given y^{-2}, rewrite the division:

$$\frac{y^2}{y^{-2}}$$

iv) Subtract the exponents:
Since division with the same base involves subtracting exponents, we have:

$$y^{2-(-2)} = y^4$$

Thus, the simplified expression for $\frac{\sqrt{y^3} \cdot \sqrt{y}}{y^{-2}}$ is: y^4